

Weekly Progress Report – Week 3

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Domain: Data Science and Machine Learning

Project Title: Forecasting of Smart City Traffic Patterns

Week Ending : 23- May-2026

I. Overview

- During Week 3 of the internship project, the focus was on developing and testing forecasting models for predicting smart city traffic patterns. After completing data preprocessing and exploratory data analysis in the previous weeks, this week's work concentrated on implementing machine learning algorithms, analyzing prediction accuracy, and visualizing forecasted traffic trends.
- The project aims to help the government improve traffic management systems by forecasting traffic congestion and understanding traffic flow behavior at different city junctions.

II. Tasks Completed

1. Machine Learning Model Development

- Implemented initial forecasting models using Python.
- Applied Linear Regression and Random Forest algorithms for traffic prediction.
- Split the dataset into training and testing datasets for model evaluation.

Trained the models using extracted traffic features such as:

- Hour
- Day of Week
- Month
- Weekend Indicator

2. Traffic Forecasting and Prediction

- Generated traffic predictions for different junctions.
- Compared actual traffic values with predicted traffic values.
- Analyzed peak traffic periods and congestion trends.
- Observed how traffic varies during weekends and working days.

3. Data Visualization

- Created graphs and visual representations for:
- Actual vs Predicted Traffic
- Hourly Traffic Trends
- Junction-wise Traffic Analysis
- Improved visualization techniques for better interpretation of forecasting results.

4. Model Evaluation

- Evaluated forecasting model performance using:
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Compared prediction accuracy between different machine learning models.
- Identified areas for model improvement and optimization.

III. Milestones Achieved

- Successfully implemented the first forecasting models for traffic prediction.
- Generated initial traffic forecasts for smart city junctions.
- Completed visualization of prediction results and traffic patterns.
- Achieved a better understanding of machine learning workflow in real-world projects.

IV. Challenges and Hurdles

1. Model Accuracy Issues

- Faced difficulty in achieving accurate traffic predictions during peak traffic hours.
- Some models produced inconsistent predictions for holidays and special events.
- Solution Implemented
- Improved feature selection and performed additional preprocessing to enhance prediction accuracy.
- Studied forecasting concepts and experimented with different parameters.

2. Understanding Machine Learning Algorithms

- Initially faced challenges in understanding how forecasting algorithms process time-series data.
- Solution Implemented
- Referred to online tutorials and documentation for better understanding of model training and evaluation techniques.
- Practiced implementing machine learning models on sample datasets.

3. Visualization Complexity

- Encountered difficulty while comparing multiple traffic trends in a single graph.
- Solution Implemented
- Used separate visualizations and clearer labeling techniques for better readability and interpretation.

V. Lessons Learned

- Learned the complete workflow of building a machine learning forecasting model.
- Improved practical knowledge of model training, testing, and evaluation techniques.
- Understood the importance of feature engineering in improving prediction performance.
- Gained hands-on experience in interpreting forecasting results and traffic behavior patterns.
- Enhanced problem-solving, analytical thinking, and Python programming skills.

VI. Next Week's Goals

- Improve forecasting model accuracy using advanced techniques.
- Explore advanced time-series forecasting methods such as ARIMA or Prophet models.
- Perform deeper analysis of traffic congestion trends and seasonal behavior.
- Prepare the final project report and presentation materials.

VII. Additional Comments

Week 3 provided valuable practical experience in machine learning model development and traffic forecasting analysis. The project continues to strengthen my understanding of predictive analytics, smart city technologies, and real-world data science applications.